

Data Analyst Internship Task 5: Exploratory Data Analysis

Author: TobyAnn Aldridge

Date: Sept. 9, 2026

Objective: Conduct Exploratory Data Analysis (EDA) to identify trends, patterns, and anomalies; use statistical auditing and data visualization to discover underlying patterns, relationships, and anomalies.

DataSet:

/kaggle/input/datasets/nabihazahid/ecommerce-dataset-for-sql-analysis/ecommerce_dataset_10000.csv

Purpose Statement:

The purpose of this report is to summarize the empirical results of an Exploratory Data Analysis conducted on an e-commerce platform dataset comprising 10,000 transaction records. This analysis evaluates continuous feature distributions, customer feedback trends, fulfillment pipeline health, and high-value basket outliers to generate data-driven business recommendations.

1. Data Integrity & Schema Audit

Prior to exploratory visualization, the dataset underwent structural profiling using `df.info()`, missingness audits, and categorical validation:

- **Dataset Shape & Non-Null Coverage:** The dataset consists of exactly 10,000 transaction rows and 24 attributes with 100% complete data integrity across all features (zero null values).
- **Entity Granularity:** The 10,000 order lines encompass 4,327 unique customers and 15 catalog items across 6 product categories.
- **Feature Engineering:** Date attributes (`order_date`, `signup_date`, `review_date`) were parsed to native datetime formats, and a line-item monetary metric was derived:

Total Revenue = Quantity x Unit Price

2. Univariate Numerical Analysis & Outlier Detection

Continuous variables were analyzed using parametric metrics, interquartile ranges (IQR), histograms, and boxplots:

Metric	Mean	Median (50%)	Std Dev	Min	Max	IQR	Distribution Profile
Quantity	3.00	3.00	1.42	1.0	5.0	2.0	Uniform discrete (1–5 units)

Unit Price	\$246.65	\$129.00	\$282.61	\$25.00	\$999.00	\$149.00	Multimodal, heavily right-skewed
Total Revenue	\$745.08	\$398.00	\$1,012.44	\$25.00	\$4,995.00	\$598.00	Exponential decay with extreme fliers
Rating	2.99	3.00	1.41	1.0	5.0	2.0	Symmetric / Uniform across 1–5 stars

Statistical Takeaways & Anomalies

- **Basket Volume:** Line-item order quantities are evenly dispersed across each integer value from 1 to 5 units (~2,000 orders per bucket) with no zero or negative quantity anomalies.
- **Pricing Skewness:** While 75% of inventory items are priced at or below \$229.00, premium catalog offerings (\$899.00 and \$999.00) plot distinctly as upper-fence outliers beyond the \$452.50 whisker boundary.
- **Revenue Outliers:** Total revenue per line item is compressed below \$796.00 for 75% of all transactions. However, multi-unit bulk purchases of high-ticket items create a continuous chain of extreme fliers stretching from \$1,800.00 to \$4,995.00, demonstrating that platform revenue is heavily leveraged by top-tier carts.

3. Categorical & Operational Distribution Profiling

Frequency evaluations using `value_counts()` and Seaborn count plots revealed specific concentration patterns:

- **Product Catalog Demand:** Platform sales are heavily concentrated in **Electronics** (2,616 orders; 26.16%) and **Apparel** (2,047 orders; 20.47%), representing nearly 47% of total sales activity. The remaining volume is evenly split across **Home & Kitchen** (13.91%), **Books** (13.34%), **Toys** (13.14%), and **Sports** (12.98%).
- **Customer Sentiment Baseline:** Ratings reflect a balanced distribution across all five tiers (~19.5% to 20.5% per star rating), averaging 2.99 / 5.0. This confirms no platform-wide bias or polarization toward extreme positive or negative sentiment.
- **Fulfillment Pipeline Health:** Orders are distributed across five operational stages: **Shipped** (20.63%), **Delivered** (20.37%), **Cancelled** (19.70%), **Pending** (19.68%), and **Returned** (19.62%). While successful fulfillment accounts for ~41% of transactions, total pipeline friction (**Cancelled** + **Returned**) impacts 39.32% of orders.
- **Demographic & Payment Uniformity:** Purchases are distributed across age demographics (**Teenagers**: 33.59%, **Senior**: 33.42%, **Adults**: 32.99%) and payment channels (**Cash on Delivery**:

34.09%, **PayPal**: 33.01%, **Credit Card**: 32.90%). Geographic demand spans 10 countries, led by France (10.60%) and China (10.54%)

4. Strategic Recommendations

1. **Target Fulfillment Friction:** With combined cancellation and return rates exceeding 39%, investigate order-handling, courier dispatching, and product expectation mismatches in the leading volume categories (**Electronics** and **Apparel**).
2. **Capitalize on Premium Carts:** High-ticket line items (\$899–\$999) generate disproportionate top-line revenue when purchased in multi-unit quantities; introduce targeted cross-selling and warranty packages for premium electronics.
3. **Regional Payment Optimization:** Given the strong preference for **Cash on Delivery** (34.09%), implement digital payment incentives to reduce transit cancellations and return-to-origin processing costs.